Dictionary

Greg was reading his favorite Computer Science textbook (no, not digitally) when he realized that the index is missing! An index helps find the page number of where a specific topic was mentioned. Since Greg doesn't want to flip through all of the pages to find the topics he's looking for, he's asking for your help.

Problem Statement

In no particular order, you are given N unique topic names where each topic name t_n can be found on page number p_n. You are then given Q queries, where each query is a topic name t_q. For each query, print the corresponding page number for t_q or print -1 if it cannot be found in the book.

Input

The first line gives the number of topic names N where $1 \leq N \leq 1000$. The next N lines each contain a page number p_n, followed by a space, followed by a topic name t_n which may contain any ASCII characters (including spaces) but does not begin or end with any whitespace characters.

The line after that contains the number of queries Q where $1 \leq Q \leq 10^5$. The next Q lines each contain a topic name t_q.

You are also given:
$1 \leq p_n \leq 10^5$
$1 \leq |t_n| \leq 1000$

Output

For each t_q, print its corresponding page number in the book or print -1 if it cannot be found.

Sample Input

```
3
23 recursion
109 graph theory
1010 binary
5
binary
prime numbers
graph theory
recursion
graph
```

Sample Output

```
1010
-1
109
23
-1
```